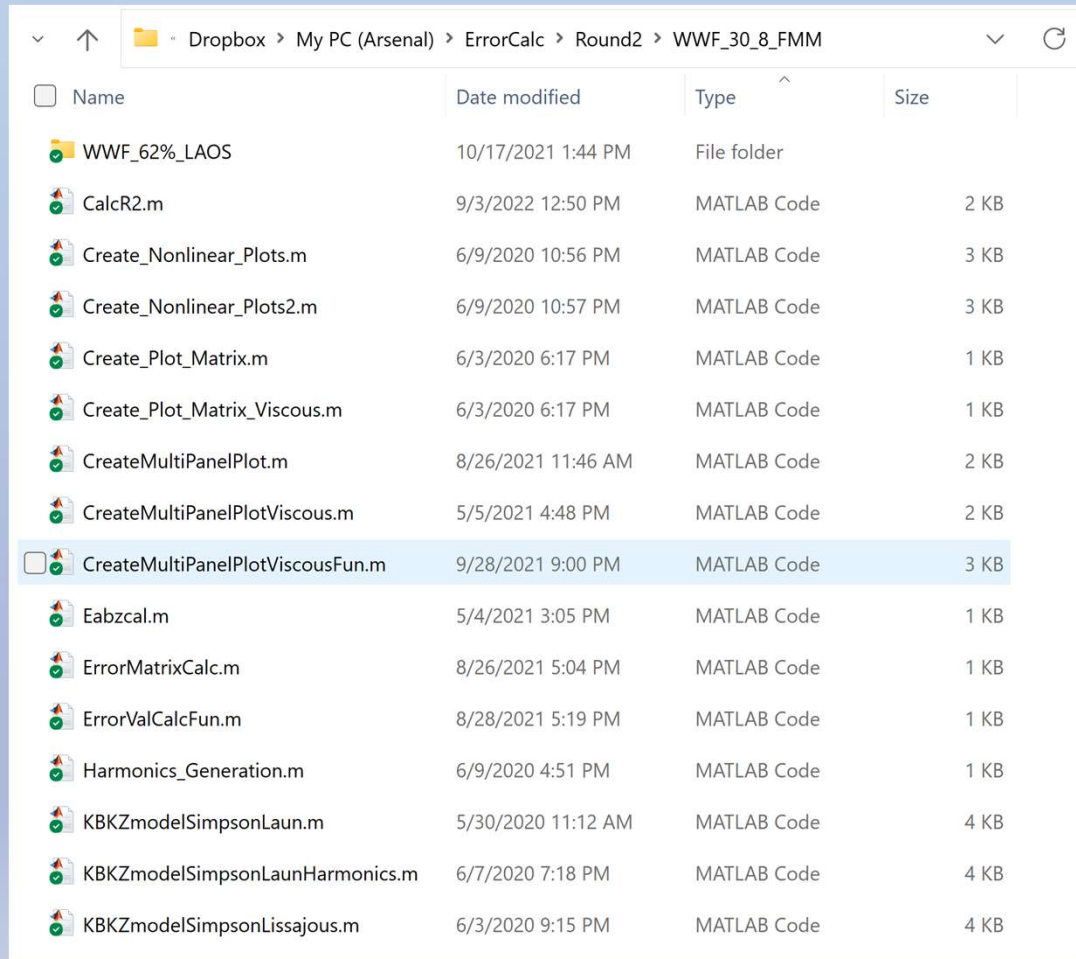


Running K-BKZ model on Matlab

Folder Structure



☐ Name	Date modified	Type	Size
WWF_62%_LAOS	10/17/2021 1:44 PM	File folder	
CalcR2.m	9/3/2022 12:50 PM	MATLAB Code	2 KB
Create_Nonlinear_Plots.m	6/9/2020 10:56 PM	MATLAB Code	3 KB
Create_Nonlinear_Plots2.m	6/9/2020 10:57 PM	MATLAB Code	3 KB
Create_Plot_Matrix.m	6/3/2020 6:17 PM	MATLAB Code	1 KB
Create_Plot_Matrix_Viscous.m	6/3/2020 6:17 PM	MATLAB Code	1 KB
CreateMultiPanelPlot.m	8/26/2021 11:46 AM	MATLAB Code	2 KB
CreateMultiPanelPlotViscous.m	5/5/2021 4:48 PM	MATLAB Code	2 KB
☐ CreateMultiPanelPlotViscousFun.m	9/28/2021 9:00 PM	MATLAB Code	3 KB
Eabzcal.m	5/4/2021 3:05 PM	MATLAB Code	1 KB
ErrorMatrixCalc.m	8/26/2021 5:04 PM	MATLAB Code	1 KB
ErrorValCalcFun.m	8/28/2021 5:19 PM	MATLAB Code	1 KB
Harmonics_Generation.m	6/9/2020 4:51 PM	MATLAB Code	1 KB
KBKZmodelSimpsonLaun.m	5/30/2020 11:12 AM	MATLAB Code	4 KB
KBKZmodelSimpsonLaunHarmonics.m	6/7/2020 7:18 PM	MATLAB Code	4 KB
KBKZmodelSimpsonLissajous.m	6/3/2020 9:15 PM	MATLAB Code	4 KB

1. Folder will be given to you in shared Teams folder
2. Create a folder for the oil water composition you have LAOS data for.
3. The sub-folder WWF_62%_LAOS shown in this screen shot contains experimental data. This is where you will place your data for the LAOS experimental data. Please change the name of subfolder and names of text files that store the experimental data as you see necessary.
4. The names of the text files must be exact. The names also have to contain the values of strain amplitude and oscillation frequency.

Experimental data in the sub-folder

- Experimental data is saved as text file.
- Each text file contains data for one LAOS result.
- The text file structure is as follows:
 - there are 5 columns
 - Column 1: strain
 - Column 2: total stress
 - Column 3: strain rate
 - Column 4: viscous contribution to stress
 - Column 5: elastic contribution to stress

Text file example

Example file:

Exp_WGF62%_25%Om1.txt

Strain amplitude
percent

Oscillation
frequency

Strain	Total Stress	Strain Rate	Viscous Stress	Elastic Stress
0	649.792	0.249006	649.792	0
0.00130376	652.461	0.249003	649.793	2.66815
0.00260748	655.132	0.248992	649.796	5.33581
0.00391112	657.803	0.248975	649.801	8.00251
0.00521466	660.475	0.248951	649.807	10.6678
0.00651806	663.146	0.248921	649.815	13.3311
0.00782128	665.818	0.248883	649.825	15.992
0.00912429	668.488	0.248839	649.837	18.6501
0.010427	671.156	0.248788	649.851	21.3049
0.0117295	673.823	0.24873	649.867	23.956
0.0130317	676.487	0.248665	649.884	26.6028
0.0143334	679.149	0.248593	649.904	29.2451
0.0156348	681.807	0.248515	649.925	31.8823
0.0169358	684.462	0.248429	649.948	34.5141
0.0182363	687.112	0.248337	649.972	37.1401
0.0195363	689.758	0.248238	649.998	39.76
0.0208358	692.4	0.248133	650.027	42.3734
0.0221347	695.036	0.24802	650.056	44.98
0.023433	697.667	0.247901	650.088	47.5794
0.0247306	700.292	0.247775	650.121	50.1715
0.0260276	702.912	0.247642	650.156	52.7559
0.0273239	705.525	0.247502	650.192	55.3324
0.0286194	708.131	0.247356	650.23	57.9008
0.0299141	710.731	0.247203	650.27	60.4608

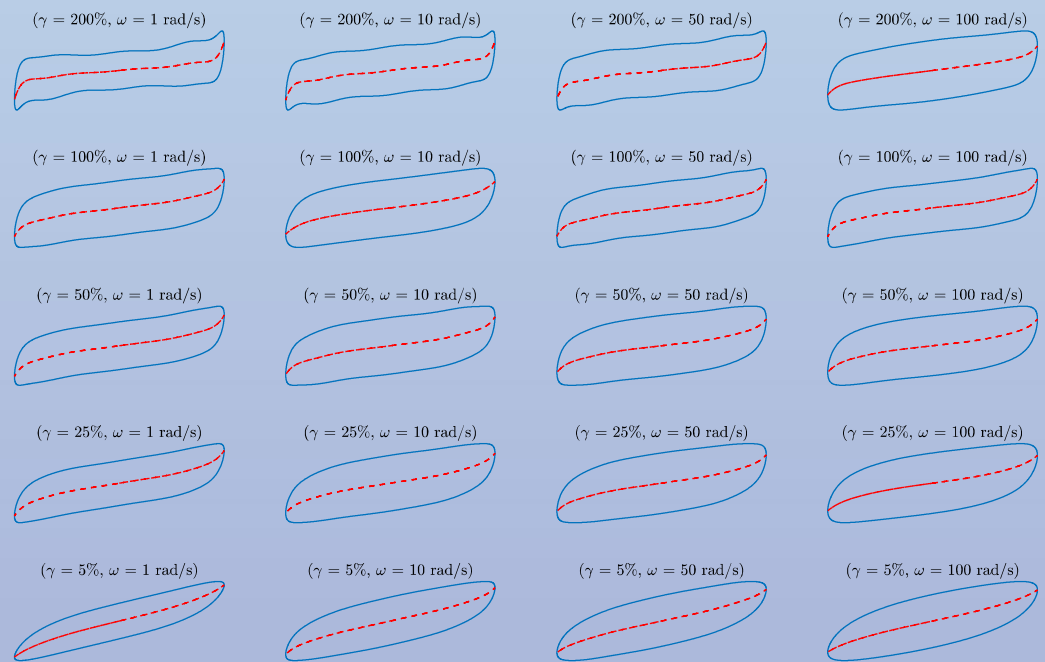
Plotting Experimental Data

- There are Matlab codes written by Eng. Isam to plot experimental data of Lissajous plots.
- Those codes are named: “CreateMultiPanelPlotElastic.m” and “CreateMultiPanelPlotViscous.m”
- To run those codes, first adjust the file path in the codes: (FilePath= ...)
- Copy file path from “file explorer”
- Open the Matlab m file. Then make sure you change the current folder to where the Matlab m file is.
- Change the base name of the text files in line 23 of the two codes as you see necessary.
- Run the Matlab code by writing the name of the Matlab file in the command window.

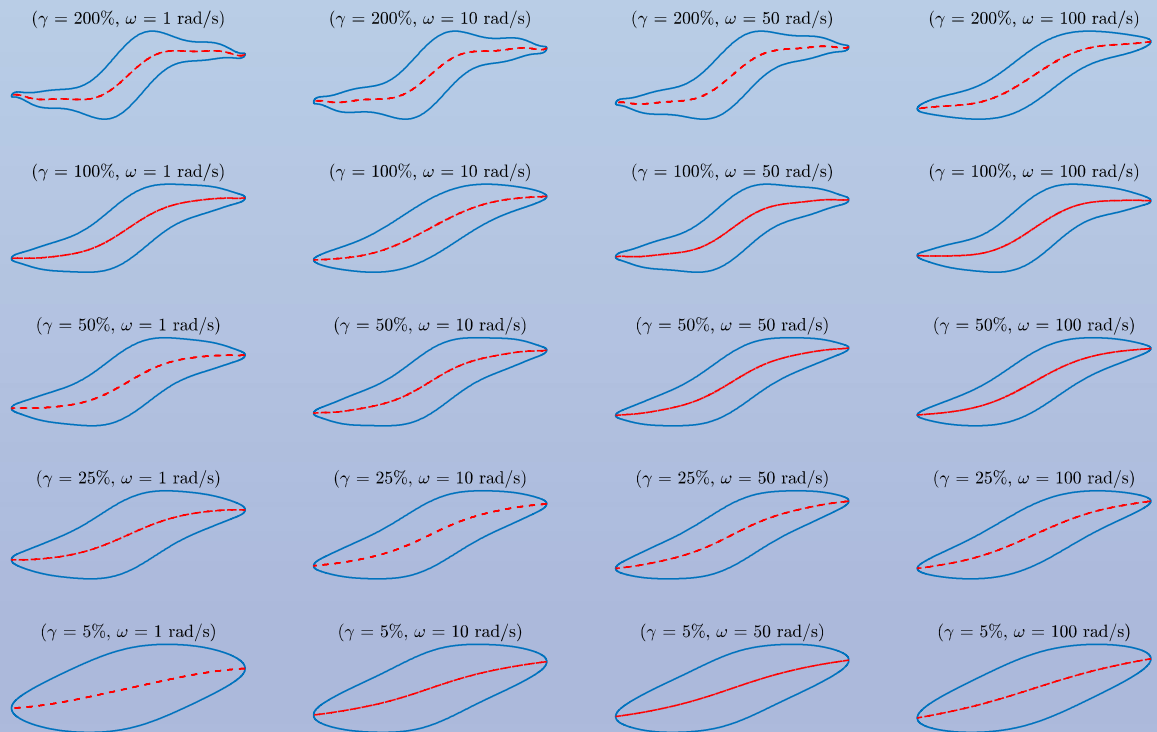
Matlab Code: “CreateMultiPanelPlotViscous.m”

```
1  FilePath=['C:\Users\wahab\Dropbox\My PC (Arsenal)\ErrorCalc\Round2\WWF_30_8_FMM\WWF_62%_LAOS\'];
2
3
4
5  omegaV = [1, 10,50, 100];
6  AmpV = [0.05, 0.25, 0.5, 1.0, 2.0];
7
8
9
10 % Create plots
11 t = tiledlayout(size(AmpV, 2),size(omegaV, 2));
12
13 for j=size(AmpV, 2):-1:1
14     for i=1:size(omegaV, 2)
15
16
17
18
19
20         omega = omegaV(i); Amp = AmpV(j);
21         AmpP = Amp*100;
22
23         FileName = strcat('Exp_WGF62%', num2str(AmpP), '%0m',num2str(omega), ...
24             '.txt' );
25
26
27
28         q = load(strcat(FilePath, FileName) );
29
30
31         nexttile;
32         plot(q(:,3)/max(q(:,3)), q(:,2)/max(q(:,2) ), 'LineWidth',0.8 );
33         hold on
34         plot(q(:,3)/max(q(:,3)), q(:,4)/max(q(:,2) ), '--r', 'LineWidth',0.8 );
35
36         title(['(\gamma$ = ',num2str(AmpP),'\%', ' $\omega$ = ',num2str(omega), ' rad/s)'], 'interpreter','latex')
37
38         axis off
39
40
41     end
42
43 end
44
45
```

```
Command Window
>> CreateMultiPanelPlotElastic
fx>> |
```



```
>> CreateMultiPanelPlotViscous  
fx>>
```



Edit and Run K-BKZ code

Procedure for editing and running K-BKZ code:

- 1- Put the newly found G_i and τ_i in the Matlab file KBKZmodelSimpsonSoskeyLissajous.m
- 2- Edit files: Lissajous_Plots_GenerationFun.m and CreateMultiPanelPlotViscousFun.m as we will detail in following slides.
- 3- In the command prompt type Lissajous_Plots_GenerationFun ([a b]), where a and b are the Winter Soskey parameters values. Once finished, type CreateMultiPanelPlotViscousFun.

Lissajous_Plots_GenerationFun.m

```
1 function Lissajous_Plots_GenerationFun(para)
2
3 aSosval = para(1);
4 bSosval = para(2);
5 omegaV = [1, 10, 50, 100];
6 AmpV = [0.05, 0.25, 0.5, 1.0, 2.0];
7
8 dim = size(omegaV, 2)*size(AmpV, 2);
9
10 parfor ij=1:dim
11     i = floor((ij-1)/size(AmpV, 2))+1;
12     j = ij-(i-1)*size(AmpV, 2);
13     disp(ij);
14     omega = omegaV(i); Amp = AmpV(j);
15     AmpP = Amp*100;
16     [time2, strain, strainRt, tau12t2, ElStr, VisStr] = ...
17         KBKZmodelSimpsonSoskeyLissajous([omega, Amp, aSosval, bSosval]);
18     disp(ij);
19     tau12t2 = ElStr+VisStr;
20     figure('Name', strcat('ElasticSt', num2str(AmpP), '%Om', num2str(omega) ));
21     plot(strain/max(strain), tau12t2/(max(tau12t2)), 'k', 'LineWidth', 1);
22     hold on
23     plot(strain/max(strain), ElStr/(max(tau12t2)), '--r', 'LineWidth', 0.8);
24     axis off
25     hold off
26     saveas(gcf, strcat('ElasticSt', num2str(AmpP), '%Om', num2str(omega) ), 'fig')
27
28
29     figure('Name', strcat('ViscousSt', num2str(AmpP), '%Om', num2str(omega) ));
30
31     data = [strain; tau12t2; strainRt; VisStr; ElStr];
32
33     dlmwrite(strcat('Theo_WGF62%', num2str(AmpP), '%Om', num2str(omega), ...
34         '.txt'), data, 'delimiter', '\t', 'newline', 'pc');
35
36     plot(strainRt/max(strainRt), tau12t2/(max(tau12t2)), 'k', 'LineWidth', 1);
37     hold on
38     plot(strainRt/max(strainRt), VisStr/(max(tau12t2)), '--r', 'LineWidth', 0.8);
39     axis off
40     hold off
41     saveas(gcf, strcat('ViscousSt', num2str(AmpP), '%Om', num2str(omega) ), 'fig');
42
43
44 end
45
46 end
```

1. The function takes input of a and b of Soskey and Winters in the following format: [a b]
2. Check omegaV values and ensure that those are the frequency values used experimentally
3. Check AmpV values as well.
4. Choose a “base” name for your oil samples instead of “WGF62%”. (line 33)
5. Keep “Theo_” unchanged.

CreateMultiPanelPlotViscousFun.m

```
1 function error = CreateMultiPanelPlotViscousFun()
2
3
4
5 FilePath=[strcat(pwd, '\WWF_62%_LAOS\')];
6
7 FilePath2 = [strcat(pwd, '\')];
8
9
10
11
12 omegaV = [1, 10, 50, 100];
13 AmpV = [0.05, 0.25, 0.5, 1.0, 2.0];
14
15
16 error = 0.0;
17 % Create plots viscous
18
19 figure;
20
21 t = tiledlayout(size(AmpV, 2), size(omegaV, 2));
22
23 for j=size(AmpV, 2):-1:1
24 for i=1:size(omegaV, 2)
25
26
27
28
29
30 omega = omegaV(i); Amp = AmpV(j);
31 AmpP = Amp*100;
32
33 FileName = strcat('Exp_WGF62%', num2str(AmpP), '%0m', num2str(omega), ...
34 '.txt' );
35
36 FileName2 = strcat('Theo_WGF62%', num2str(AmpP), '%0m', num2str(omega), ...
37 '.txt' );
38
39 q = load(strcat(FilePath, FileName) );
40 qtheo = load(strcat(FilePath2, FileName2) );
41
42 nexttile;
43 plot(q(:,3)/max(q(:,3)), q(:,2)/max(q(:,2)), 'LineWidth', 0.8);
44 hold on
45 plot(q(:,3)/max(q(:,3)), q(:,4)/max(q(:,2)), '--r', 'LineWidth', 0.8);
```

```
45 plot(q(:,3)/max(q(:,3)), q(:,4)/max(q(:,2)), '--r', 'LineWidth', 0.8);
46
47 plot(qtheo(:,3)/max(qtheo(:,3)), qtheo(:,2)/max(qtheo(:,2)) ...
48 , 'g', 'LineWidth', 0.8);
49 hold on
50 plot(qtheo(:,3)/max(qtheo(:,3)), qtheo(:,4)/max(qtheo(:,2)) ...
51 , '--g', 'LineWidth', 0.8);
52
53 axis off
54
55
56 xinterp = [1:1:size(q, 1)];
57 theoElStrvals = (interp1([0:1:size(qtheo)-1], qtheo(:, 5) ./ ...
58 max(qtheo(:, 2)), (xinterp-1).*(size(qtheo,1)-1)/size(q,1) ));
59 error = error+ sum((theoElStrvals-q(:, 5)./ max(q(:,2)))^2)/size(q,1);
60
61 theoVisStrvals = (interp1([0:1:size(qtheo)-1], qtheo(:, 4) ./ ...
62 max(qtheo(:, 2)), (xinterp-1).*(size(qtheo,1)-1)/size(q,1) ));
63
64 error = error+ sum((theoVisStrvals-q(:, 4)./ max(q(:,2)))^2)/size(q,1);
65
66
67 end
68 end
69
70
71 % create plots elastic
72
73 % Create plots
74
75 figure;
76
77 t = tiledlayout(size(AmpV, 2), size(omegaV, 2));
78
79 for j=size(AmpV, 2):-1:1
80 for i=1:size(omegaV, 2)
81
82
83
84
85
86 omega = omegaV(i); Amp = AmpV(j);
87 AmpP = Amp*100;
```

```
88
89 FileName = strcat('Exp_WGF62%', num2str(AmpP), '%0m', num2str(omega), ...
90 '.txt' );
91
92 FileName2 = strcat('Theo_WGF62%', num2str(AmpP), '%0m', num2str(omega), ...
93 '.txt' );
94
95 q = load(strcat(FilePath, FileName) );
96 qtheo = load(strcat(FilePath2, FileName2) );
97
98 nexttile;
99 plot(q(:,1)/max(q(:,1)), q(:,2)/max(q(:,2)), 'LineWidth', 0.8);
100 hold on
101 plot(q(:,1)/max(q(:,1)), q(:,5)/max(q(:,2)), '--r', 'LineWidth', 0.8);
102
103 plot(qtheo(:,1)/max(qtheo(:,1)), qtheo(:,2)/max(qtheo(:,2)) ...
104 , 'g', 'LineWidth', 0.8);
105 hold on
106 plot(qtheo(:,1)/max(qtheo(:,1)), qtheo(:,5)/max(qtheo(:,2)) ...
107 , '--g', 'LineWidth', 0.8);
108
109 axis off
110
111
112 end
113 end
114
115 error = error/size(AmpV, 2)/size(omegaV, 2)/2;
116 % disp(['error = ', num2str(sqrt(error))]);
117
118 end
```

CreateMultiPanelPlotViscousFun . m

1. Change file name '\WWF_62%_LAOS\' (which is the sub-folder where experimental data is stored) as necessary.
2. Change omegaV and ampV if necessary.
3. Change prefix in file name in line 33 ('WGF62%_')
4. Change prefix in file name in line 36 ('WGF62%_')
5. Change prefix in file name in line 89 ('WGF62%_')
6. Change prefix in file name in line 92 ('WGF62%_')